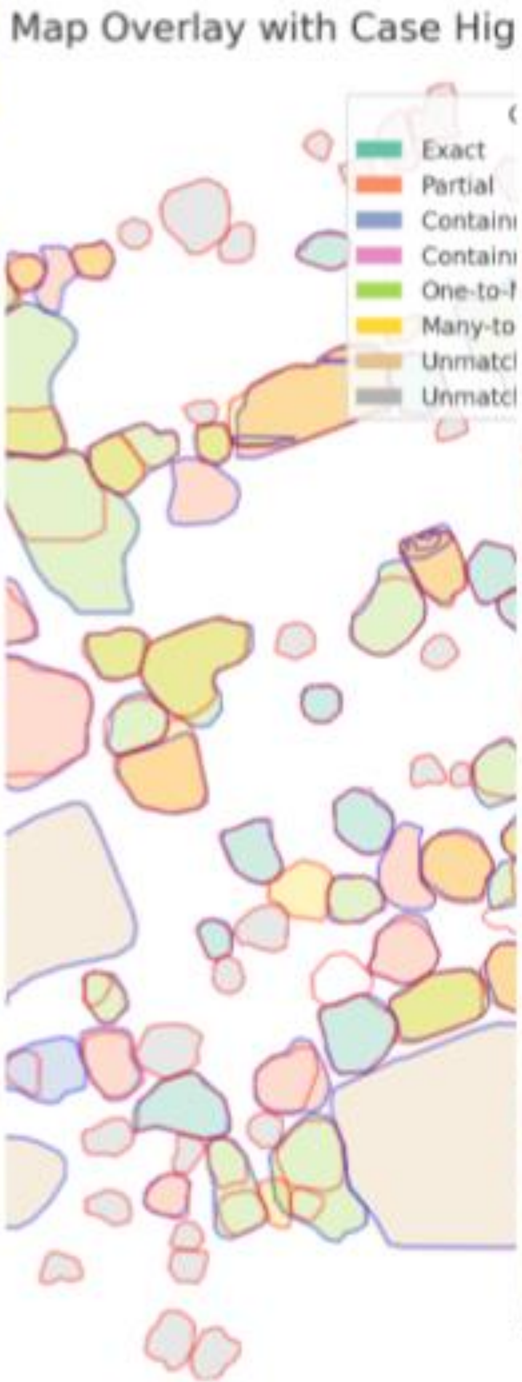


Drone Based monitoring of Tree Cover

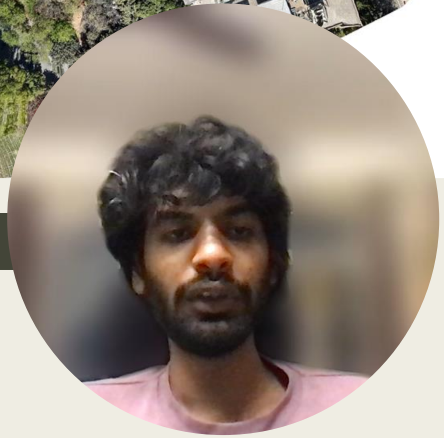
Parth Thakur • 2021CS50615

Gaurav Rajput • 2024CSY7558

Prof. Aaditeshwar Seth

Overview

- Goal: Develop an automated system for monitoring tree cover using drones
- Capture aerial images
- Generate orthomosaics
- Detect trees
- Track phenological changes.



Work Done Previously

Drone Acquisition

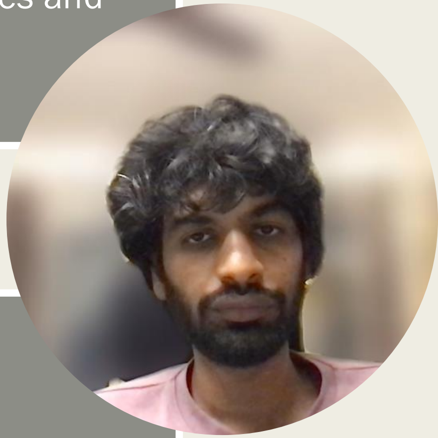
- Flight planning and on campus survey routes
- Repeated surveys of SAC, LHC, SIT
- Data management and naming convention

WebODM processing

- Orthomosaics generated for each survey date
- Georeferenced

Ground truth prepared

- CVAT polygon labels for crowns
- Exported as shapefiles and geojson



Detection baseline established

- Detectree2 polygons tuned for our site
- Tiling, buffer, threshold, simplify settings fixed

Detections over time organized

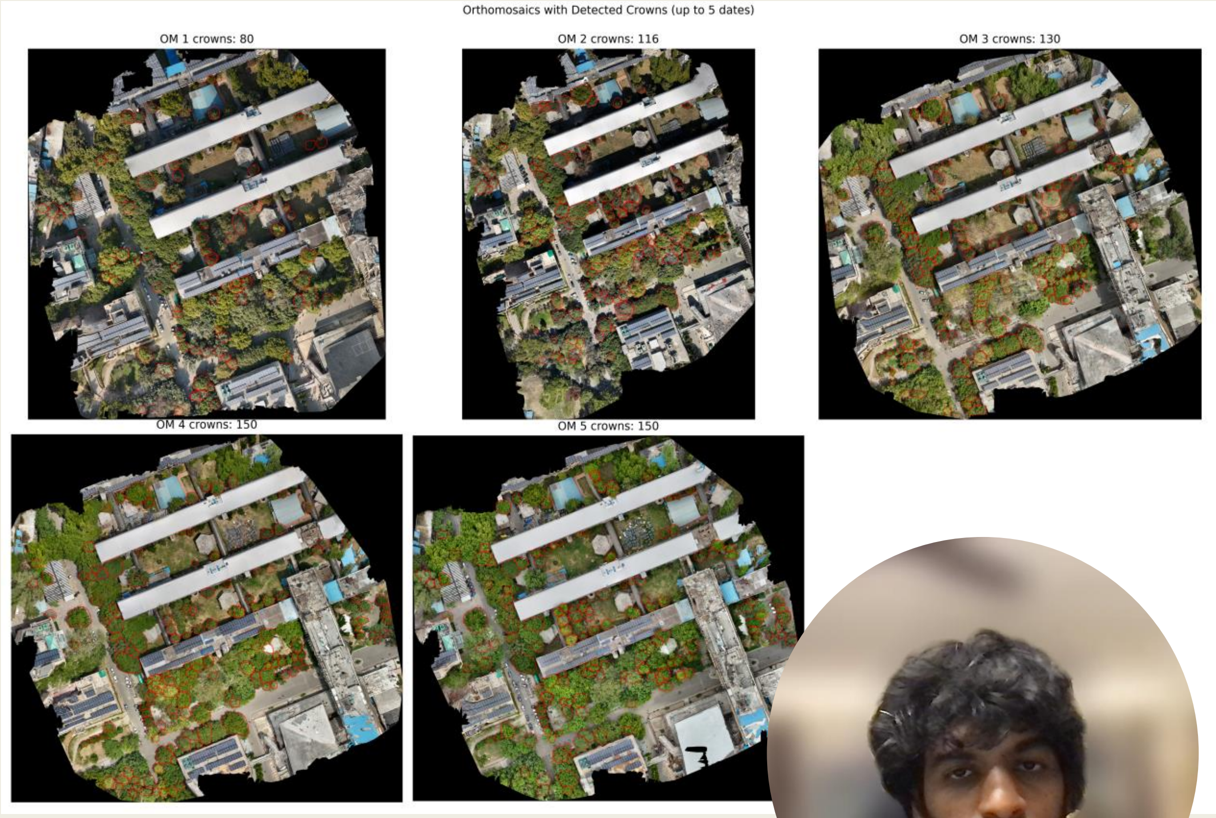
- Consistent tile size and scale across dates
- Exports stored by date and area

Now

- Tracking individual trees across orthomosaics

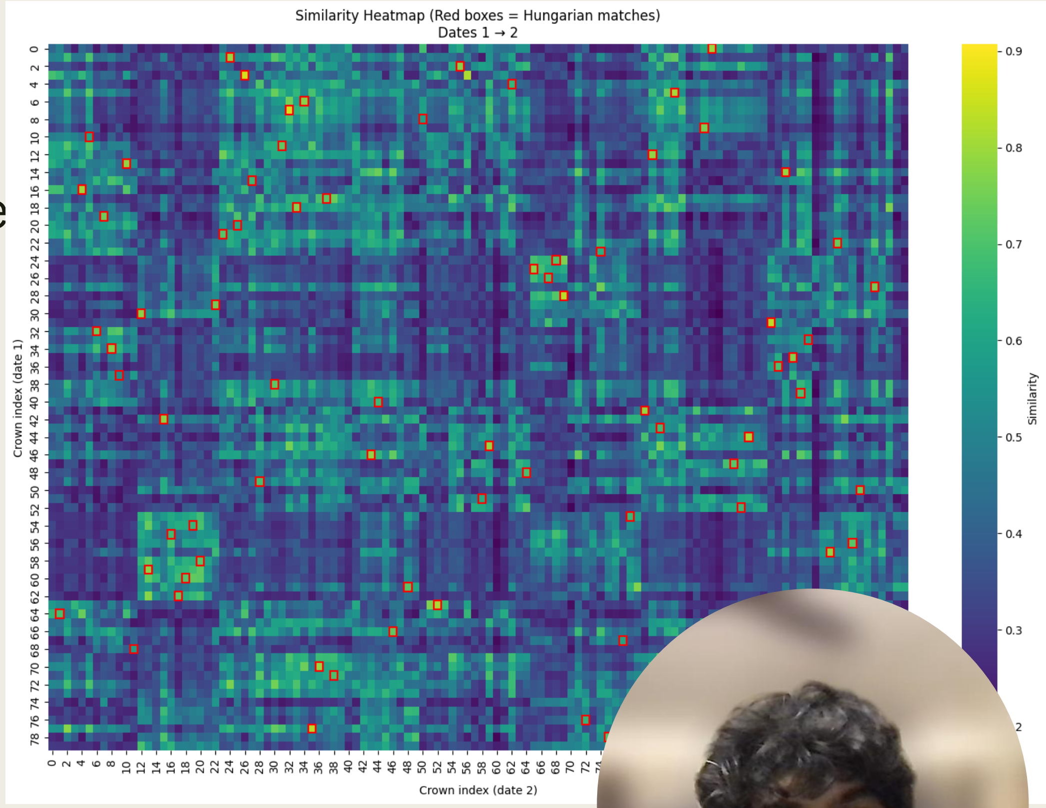
Tracking Trees Across Orthomosaics

- We Have
  - *Sequence of orthomosaics (same area, different dates)*
  - *Per-date crown polygons from Detectree2*
- The goal is to be able identify which polygons across dates represent the same physical trees to measure real phenological change.
- Why its difficult:
  - *Geolocation drift, Detection variation (lighting, shadows, season, wind), Polygon inconsistency (splits, merges, misses), Real biological change (growth, pruning, leaf/flower phases)*



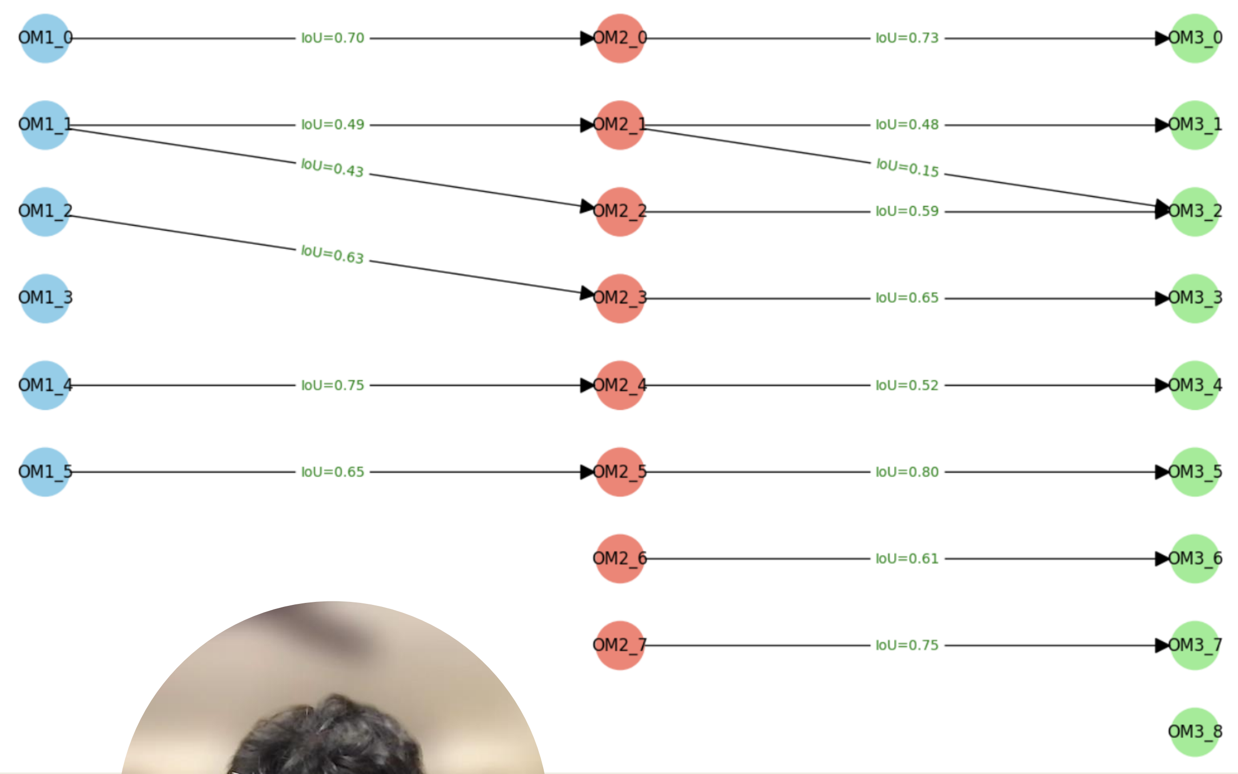
Initial approach: Hungarian Matching

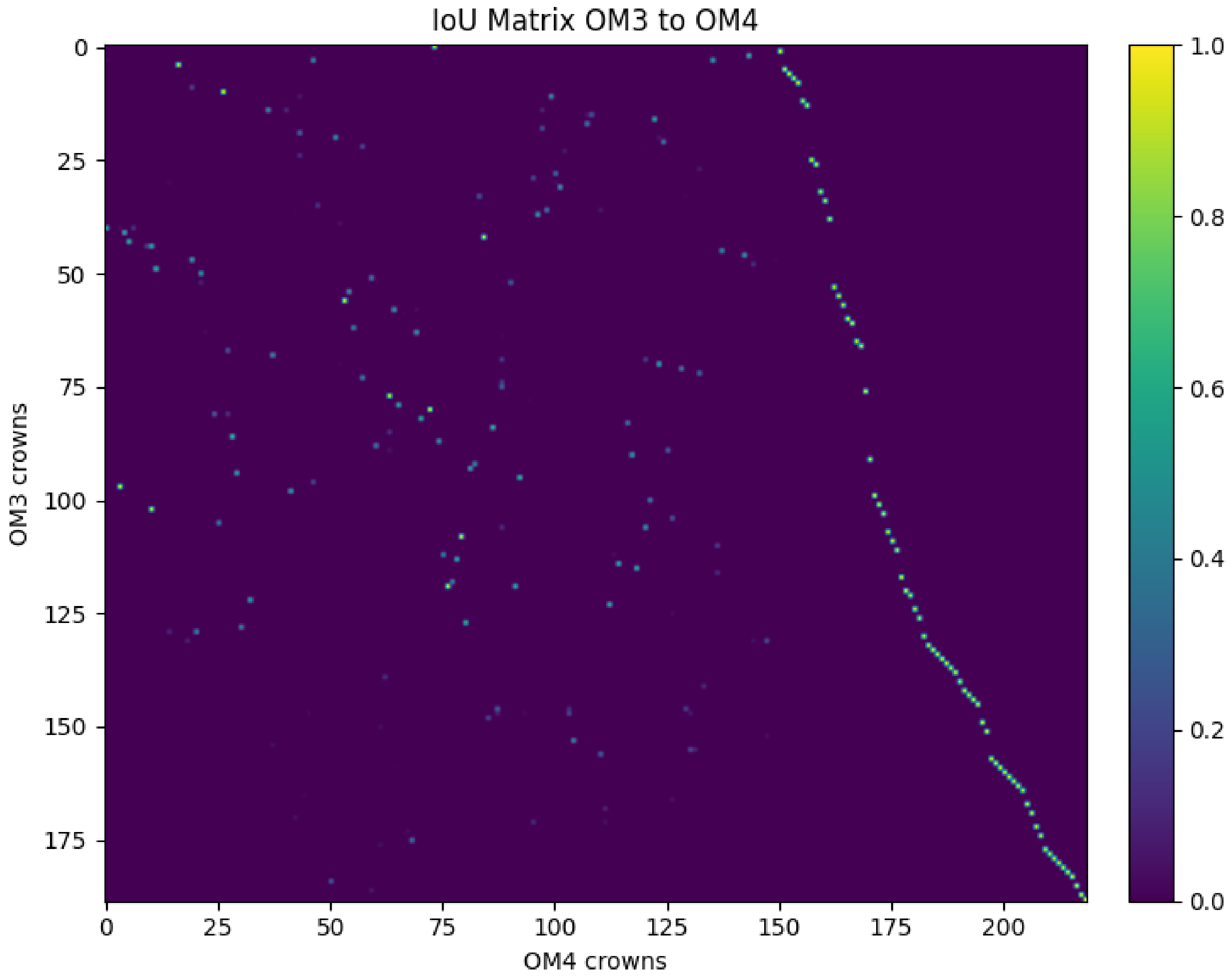
- Classic assignment algorithm (solves optimal one-to-one pairing between two sets by maximizing total similarity / minimizing cost).
- For each consecutive date: build similarity (or cost) matrix (crowns at t vs t+1), run Hungarian ⟶ get a single best partner per crown.
- Problems:
  - *One-to-one only: cannot express splits (one crown ⟶ many), merges (many ⟶ one), or temporary over-/under-segmentation.*
  - *Forced exclusivity: a slightly better similarity can “steal” a node, blocking a plausible secondary link.*
  - *Local horizon: optimizes each date pair independently—no multi-date consistency check, we’d like to keep plausible continuations for later disambiguation*



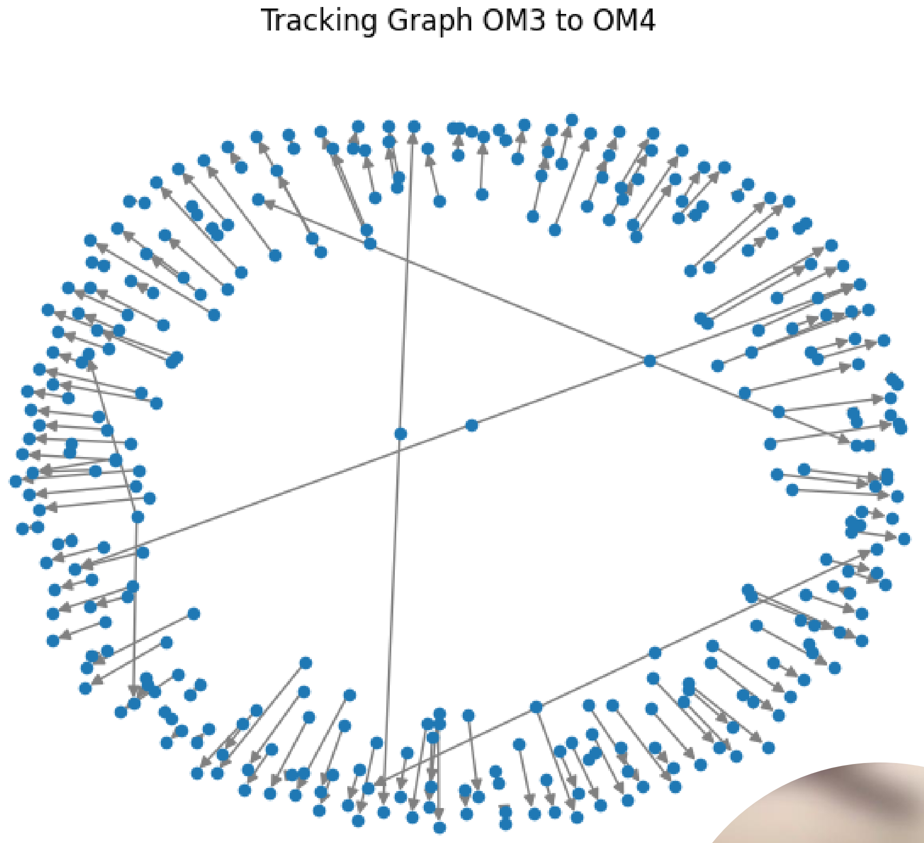
Graph Based Tracking

- Build a directed temporal graph across dates (layers = survey dates).
- Each node = (Date / OM, Crown index) with geometry + attributes (centroid, area, compactness, eccentricity, IoU-ready shape) and image patch (for appearance features).
- Edges connect crowns in consecutive dates when similarity ≥ threshold. One node may have multiple outgoing and incoming edges.
- Chains: follow best outgoing edge(s) later for phenology analysis.

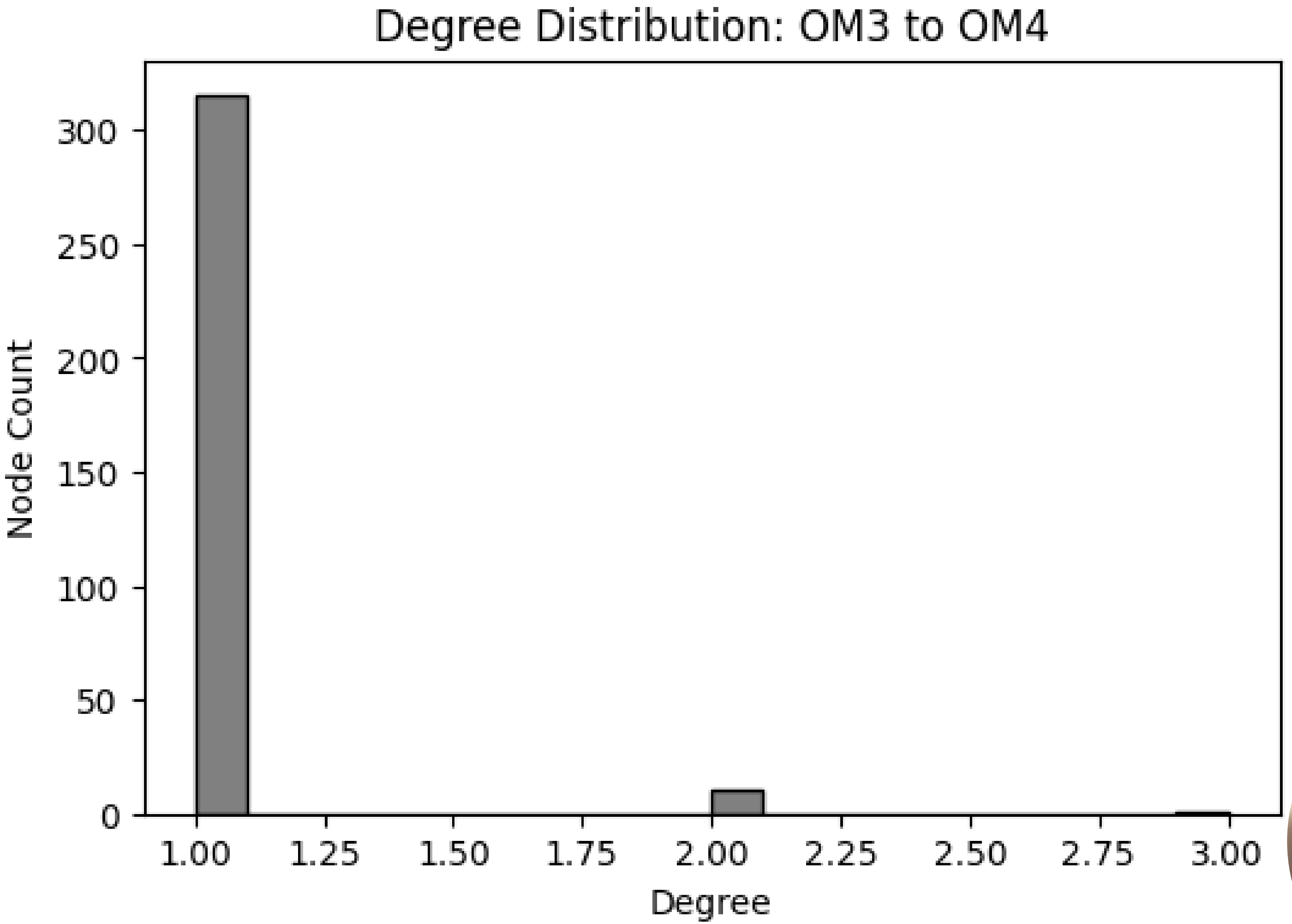




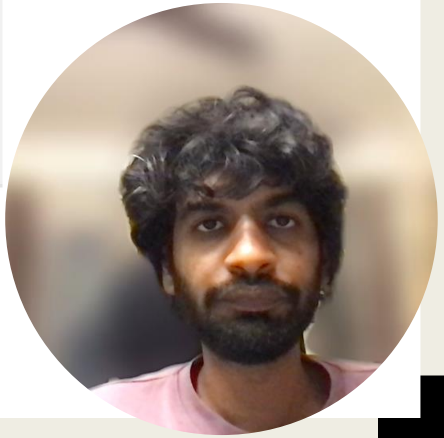
IoU Matrix OM3 to OM4
OM3 crowns
OM4 crowns
0
25
50
75
100
125
150
175
200
1.0
0.8
0.6
0.4
0.2
0.0

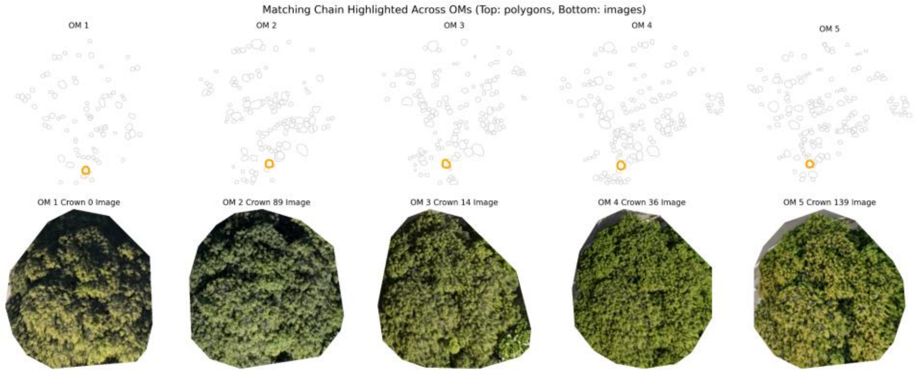


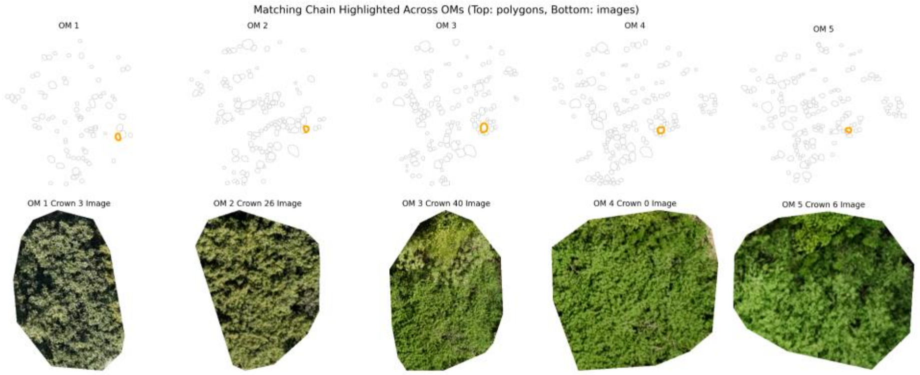
Tracking Graph OM3 to OM4

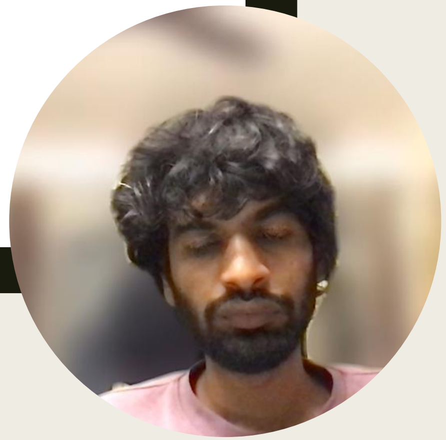


Degree Distribution: OM3 to OM4
Node Count
0
50
100
150
200
250
300
Degree
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00



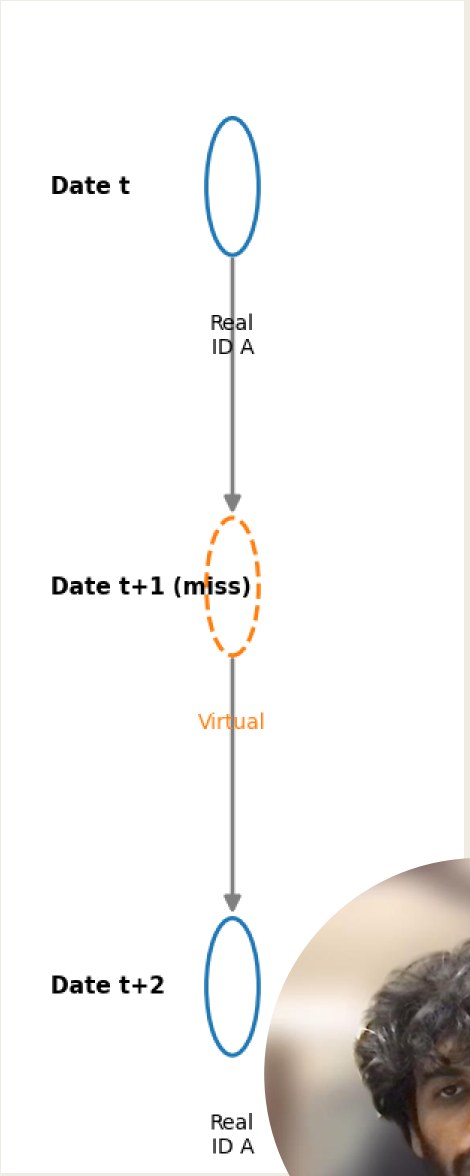

Matching Chain Highlighted Across OMs (Top: polygons, Bottom: images)
OM 1
OM 2
OM 3
OM 4
OM 5
OM 1 Crown 0 Image
OM 2 Crown 89 Image
OM 3 Crown 14 Image
OM 4 Crown 36 Image
OM 5 Crown 139 Image


Matching Chain Highlighted Across OMs (Top: polygons, Bottom: images)
OM 1
OM 2
OM 3
OM 4
OM 5
OM 1 Crown 3 Image
OM 2 Crown 26 Image
OM 3 Crown 40 Image
OM 4 Crown 0 Image
OM 5 Crown 6 Image

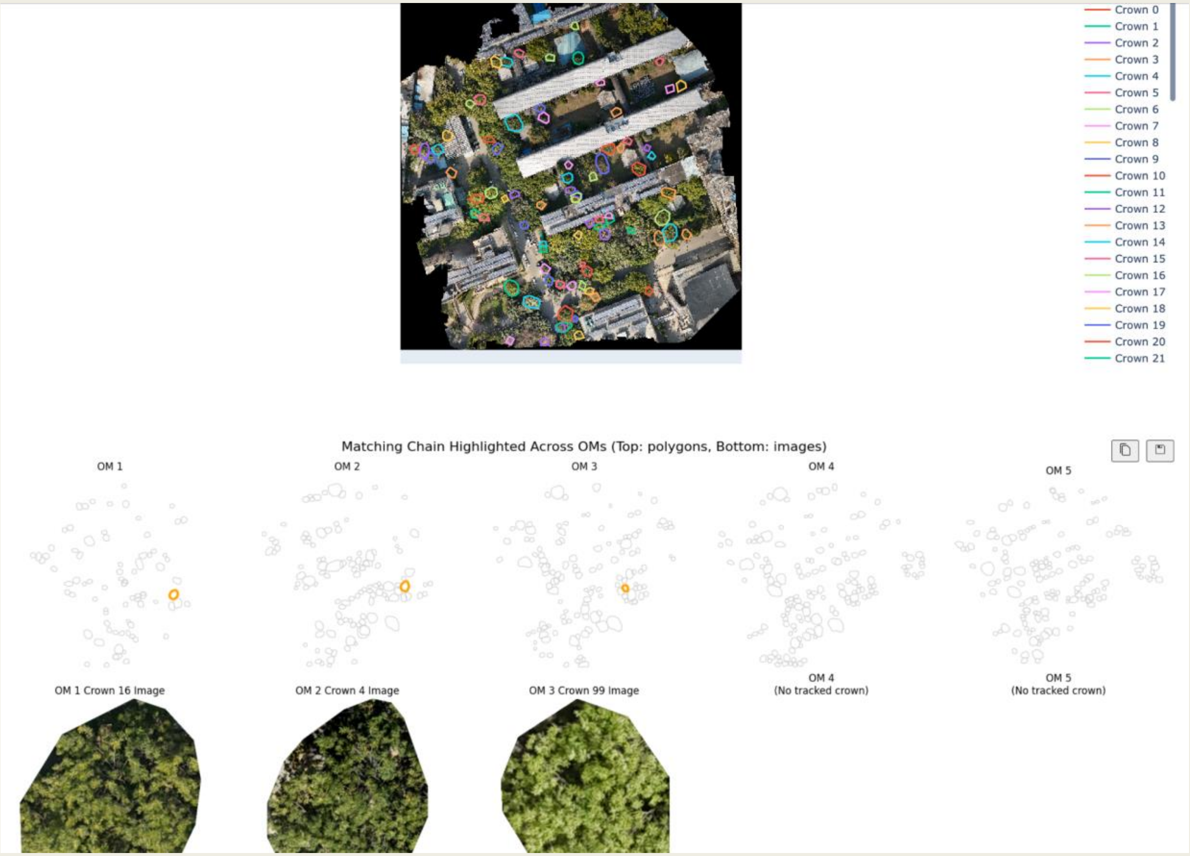


Virtual Nodes

- A real crown may be undetected in one survey → chain breaks, false “disappearance.”
- Insert a placeholder (virtual node) for that date to keep continuity.
- Node with flag virtual=True, no new polygon; attributes propagated / interpolated (centroid, area, shape)
- Maintains chain length and phenology time series. Preserve continuity when detection is imperfect.

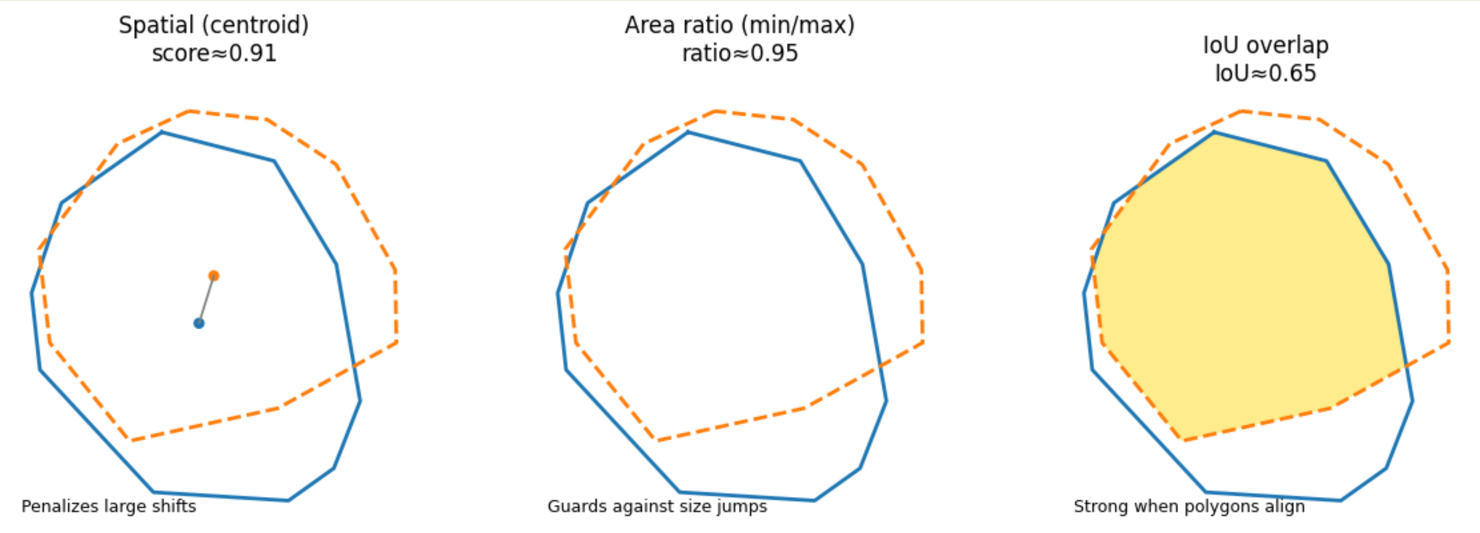


TREE-CLICKABLE TRACKING VISUALIZATION



Similarity Score

- Raw centroid distance or IoU alone is unstable (drift, detection noise).
- Combines multiple weak signals → more robust matching and keeps plausible alternatives.
- Weighted Combination of:
  - *Spatial (centroid proximity) – penalizes large shifts.*
  - *Area ratio (min/ max) – guards against drastic size changes.*
  - *IoU (overlap) – strong when polygons align; tolerant of minor drift.*



Matching Quality Metrics

- Nodes: 626
- Edges: 798
- Avg out-degree: 1.275
- Avg in-degree: 1.275
- Average diameter: 1.347
- Max Diameter: 13

Future Work

- Conditional Thresholding: Deciding the threshold based on case by case basis on the cases identified(splits, partial overlap, containment, etc.)
- Phenological monitoring of factors such as shedding, leaf coloring, and flowering based on the confident chains

THANK YOU

Questions?